

Architecture Comparison: Diffusion vs ACT

PushT Diffusion

UNet 263M params

Conditional denoising

chunk_size = 1

→ Lightweight compute / step

Speed
~13 step/s

VS

SO-101 ACT

Transformer Encoder 4L

+ CVAE Decoder 7L

chunk_size = 100 ($O(n^2)$)

→ Heavy compute / step

→ batch_size cap at 128

~1.9 step/s

Root cause: ACT chunk_size=100 Transformer self-attention $O(n^2)$ → 79GB VRAM per step at batch=128